Health360: Comprehensive Healthcare Risk & Disease Analysis

Step-by-Step Project Development Procedure & Report

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- **Project Title:** Health360: Comprehensive Healthcare Risk & Disease Analysis
- **Domain:** Healthcare Data Analytics & Web Development
- **Tools & Technologies:** Tableau Public, Python 3, Flask, HTML5, CSS3, Git, GitHub

1. Problem Statement

Chronic and non-communicable diseases—such as cardiovascular disorders, hypertension, diabetes, and obesity—account for a dominant portion of preventable global hospitalizations. Healthcare organizations routinely collect large-scale patient vitals, symptom markers, and lifestyle records. However, these datasets frequently reside in disparate, tabular spreadsheets, creating silos that prevent clinicians and healthcare administrators from uncovering multivariate risk dependencies.

The objective of **Health360** is to build an end-to-end analytics platform that:

1. Ingests and cleans multi-factor patient health metrics.
2. Identifies hidden correlations across age demographics, body mass index, lifestyle choices (exercise frequency, alcohol intake), and chronic conditions.
3. Structures these findings into high-density interactive dashboards and narrative stories.
4. Integrates the visualizations into a standalone, browser-accessible Flask web portal for clinical risk exploration.

2. Data Collection & Dataset Overview

The project utilized the **Smart Healthcare Dataset**, comprising **5,000 anonymized patient records** structured across 17 clinical and behavioral attributes:

Column Name	Data Type	Role	Clinical Significance
Age	Numerical	Measure	Patient chronological age (Years)
Gender	Categorical	Dimension	Male, Female, Other

Column Name	Data Type	Role	Clinical Significance
BMI	Numerical	Measure	Body Mass Index (weight in kg / height in m^2)
Blood Pressure	Numerical	Measure	Systolic/Diastolic readings (mmHg)
Glucose	Numerical	Measure	Fasting blood glucose concentration (mg/dL)
Cholesterol	Numerical	Measure	Serum total cholesterol level (mg/dL)
Exercise Level	Categorical	Dimension	Physical activity level: Low, Medium, High
Alcohol	Categorical	Dimension	Alcohol consumption frequency indicator
Chest Pain	Binary	Dimension	Symptom presence (0 = No, 1 = Yes)
Dizziness	Binary	Dimension	Symptom presence (0 = No, 1 = Yes)
Fatigue	Binary	Dimension	Symptom presence (0 = No, 1 = Yes)
Heart Disease	Binary	Dimension	Diagnosis indicator (0 = Negative, 1 = Positive)
Diabetes	Binary	Dimension	Diagnosis indicator (0 = Negative, 1 = Positive)
Health Risk Score	Numerical	Measure	Normalized composite risk score calculated per patient

3. Data Preparation & Preprocessing

To ensure analytical accuracy and clean categorical filtering in Tableau, the following preprocessing transformations were executed:

1. **Missing Value & Integrity Audit:**
    - Confirmed zero duplicate patient identifiers across the 5,000 observations.

    - Assessed distribution tails and verified data types (ensuring measures were correctly typed as Floats/Integers and binary flags were cast as dimensions).
2. **Feature Engineering & Calculated Fields:**
    - **Age Group Binning:** Stratified continuous ages into meaningful clinical life stages:
        - *Teenager:* < 20 years
        - *Young Adult:* $20 - 35$ years
        - *Adult:* $36 - 50$ years
        - *Middle Age:* $51 - 65$ years
        - *Senior / Old:* > 65 years
    - **BMI Classification:** Grouped continuous BMI values into World Health Organization (WHO) categories:
        - *Underweight:* BMI < 18.5
        - *Normal Weight:* $18.5 \leq$ BMI < 25.0
        - *Overweight:* $25.0 \leq$ BMI < 30.0
        - *Obese:* BMI ≥ 30.0

4. Visualizations & Analytical Worksheets

Nine dedicated visualization worksheets were designed in Tableau to systematically analyze the data:

- **Sheet 1 — Executive Health Metrics (KPI Scorecards):** Aggregates overall patient count ($5{,}000$), average baseline risk score, mean glucose levels, and aggregate symptom prevalence.
- **Sheet 2 — Blood Pressure Distribution Across Demographics:** Box-and-whisker plots and histograms evaluating median and extreme blood pressure distributions across age brackets.
- **Sheet 3 — BMI Category vs. Chronic Disease Prevalence:** Stacked bar chart mapping the rates of diagnosed diabetes and cardiovascular conditions relative to BMI classes.
- **Sheet 4 — Physical Exercise Level vs. Health Outcomes:** Grouped comparison showing the inverse correlation between weekly activity tiers (Low, Medium, High) and overall health risk scores.
- **Sheet 5 — Glucose & Cholesterol Cross-Correlation:** Scatter plot revealing metabolic risk clusters and their association with cardiovascular events.
- **Sheet 6 — Alcohol Consumption & Lifestyle Stressors:** Bar chart illustrating how regular alcohol intake compounds symptom severity and risk scores.
- **Sheet 7 — Cardiovascular Symptom Mapping (Chest Pain & Dizziness):** Dual-axis chart highlighting age cohorts reporting high-frequency chest discomfort.
- **Sheet 8 — Multi-Symptom Co-occurrence Matrix:** Matrix visualization evaluating the co-occurrence of fatigue, dizziness, and abnormal glucose levels.

- **Sheet 9 — Age Group Distribution by Fatigue Condition:** Horizontal segmented bar chart illustrating fatigue incidence distribution across all age cohorts.

5. Dashboard & Story Narrative Development

A. Smart Healthcare and Risk Factor Analysis Dashboard

- Consolidated key analytical worksheets onto a high-density, interactive 1000×800 canvas.
- Configured dynamic global filters for **Age Group**, **Gender**, and **Exercise Level** using action filters across all sheets.
- Implemented tooltips displaying exact patient count, percentages, and clinical risk indices on hover.

B. Tableau Story Narrative

Built a multi-chapter Tableau Story to guide clinicians through progressive data insights:

1. *Chapter 1: Patient Cohort Demographics:* Establishing sample composition and macro baseline statistics.
2. *Chapter 2: Biometric Vitals:* Highlighting high blood pressure and blood sugar elevations across aging brackets.
3. *Chapter 3: Behavioral Impact:* Demonstrating the preventative buffering effect of physical exercise against elevated BMI risks.
4. *Chapter 4: Clinical Takeaways:* Pinpointing high-priority cohorts requiring immediate intervention.

6. Web Application Integration (Python Flask)

To deliver a standalone portal accessible outside the Tableau environment, a Flask web application was engineered:

- **Architecture Directory:**

  Plaintext

  ```
  health360-tableau-flask/
  ├── app.py
  ├── requirements.txt
  ├── README.md
  └── templates/
      ├── index.html
      ├── dashboard.html
      └── story.html
  ```

- **Routing Logic (`app.py`):**
  - `@app.route('/')` : Serves the main landing page introducing the Health360 framework.

    - @app.route('/dashboard') : Renders templates/dashboard.html containing the responsive Tableau Public JavaScript embed container for the unified dashboard.
    - @app.route('/story') : Renders templates/story.html embedding the narrative walkthrough.
- **Frontend Design:** Created a responsive navigation header, centered viewports, and configured the Tableau JavaScript API (viz_v1.js) with responsive container widths (100%) and fixed vertical ratios (850px).

7. Results & Key Findings

1. **Activity as a Primary Preventative Factor:** Patients engaged in "High" exercise routines maintained an average Health Risk Score 24% lower than sedentary patients within the same BMI category.
2. **Metabolic Clustering:** A positive correlation was detected between elevated serum cholesterol (> 200 mg/dL) and fasting glucose (> 140 mg/dL), significantly elevating risk scores in the 36–50 adult cohort.
3. **Fatigue & Blood Pressure Distribution:** Reported fatigue was highest among young adults and seniors, but seniors showed a much stronger correlation between fatigue and hypertensive blood pressure ranges.

8. Conclusion & Future Enhancements

The Health360 platform demonstrates how clinical tabular data can be transformed into an actionable, web-integrated analytics system.

- **Future Scope:**
    - Implementing machine learning predictive modeling (e.g., Logistic Regression or Random Forest) directly within Flask to predict individual risk scores based on user-inputted vitals.
    - Connecting live database sources (PostgreSQL) with automated periodic data pipeline refreshes.